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学生学号： 42353012

学生姓名： 许哲圣

学 院： 特拉华数据科学学院

年级专业： 2023 级 金融数学

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Capability Surprise and the Pricing of an Early-Commercial-Stage Foundation-Model Lab: Evidence from Zhipu AI (2513.HK)

GitHub Repository: <https://github.com/ericxuzhesheng/Zhipu-AI-Valuation>

Abstract

In January 2026, Zhipu AI (2513.HK) became the first publicly traded foundation-model laboratory. By 31 August its shares were 10.3 times the IPO price, implying US\$71.3bn of equity value. Results released after that close showed H1 revenue of RMB 953.9m, up 399.7%; API revenue contributed 86.5%, while operating loss widened to RMB 2.15bn. We distinguish those disclosures from forecast choices and apply scenario DCF, reverse DCF and layered comparables. On a US\$700m FY2026E revenue estimate, the DCF spans HK\$25–233 per share and gives a probability-weighted HK\$88, or 7.4% of the pre-results close. Private transactions imply HK\$240–457; the closest listed peer implies HK\$332 on matched LTM revenue. The reverse DCF requires about US\$92bn of 2035 revenue. Replacing earnings surprises with *capability-surprise* events then shows how model progress is priced. The evidence suggests that public investors attach a large, fragile right-tail value to future commercialisation.

Keywords: foundation models; Chapter 18C; pre-profit valuation; real options; market efficiency; capability surprise; PEAD/PCAD.

1 Introduction

The first half of 2026 produced an event with no precedent in capital-market history: within forty-eight hours, two Chinese “AI tiger” laboratories—Zhipu AI (2513.HK, 8 January) and MiniMax (00100.HK, 9 January)—became the first foundation-model developers to go public anywhere, listing on the Hong Kong Stock Exchange. Zhipu listed under Chapter 18C, the regime for specialist-technology firms that admits pre-profit issuers. At the 31 August close its shares were 10.3 times the IPO price, giving a post-placement market capitalisation of US\$71.3bn. Against US\$209.5m of LTM revenue through 2026H1, that is an equity-value-to-revenue multiple of $340.5 \times^{1,3,6,9}$. The broader frontier-AI financing wave has therefore moved from private markets into public price discovery.

This paper poses three questions: **(i)** what is Zhipu worth on fundamentals; **(ii)** what does the re-rating imply—efficient pricing of capability, exuberance, or option value; and **(iii)** is the open-weight, cost-disruption strategy sustainable? Our contribution is methodological: because an early-stage lab has no earnings, the standard earnings-surprise event study^{10,11} has nothing to bite on, so we reframe the information event as a *capability surprise*—a model release or benchmark jump—adapting the surprise-event (*chao yuqi*) design of Guosen Securities¹² to a new asset class. The weight of the paper falls on the valuation (Section 4) and the event study (Section 5); Sections 2–3 give only the context they require.

2 Background: Company, Financials, and Listing

Business model. Founded in 2019 out of Tsinghua University’s Knowledge Engineering Group, Zhipu develops the GLM (“General Language Model”) family and monetises through usage-based API access, enterprise subscriptions, and private/on-premise deployment. Its strategic choice is an *open-weight funnel*: flagship models are released with open weights and aggressive token pricing, converting developer mindshare into a usage flywheel via the Z.ai and Z.code platforms.

Financials (full statements in Appendix A). The 31 August interim-results announcement materially changes the starting point. H1 revenue rose 399.7% to RMB 953.9m; open-platform and API revenue reached RMB 825.2m, or 86.5% of the total, and its gross margin improved from -0.4% to 24.6% . The mix shift diluted total gross margin from 50.0% to 26.4% . Operating loss widened 13.0% to RMB 2.15bn, and adjusted net loss widened 12.1% to RMB 1.96bn. Reported net loss narrowed as gross profit rose and losses on investor financial instruments fell sharply after listing; the adjusted measure shows that the underlying cost base did not improve at the same pace. At 30 June the group had positive equity of RMB 4.43bn, cash and short-term investments of RMB 4.50bn, and bank loans plus leases of RMB 2.60bn⁵. The announcement presents no operating-cash-flow statement. Zhipu remains *early-commercial-stage and pre-profit*; growth, API unit economics and cash runway are more informative than earnings multiples (Figure 1).

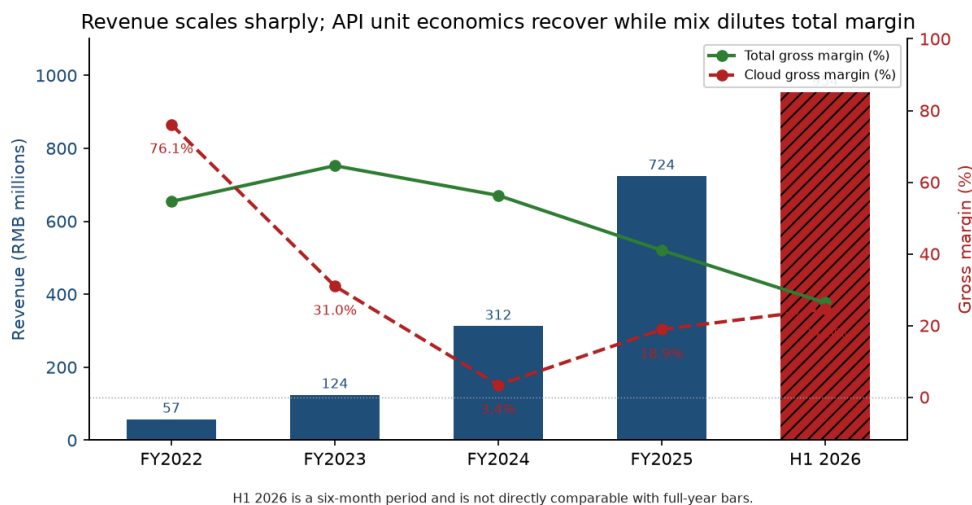


Figure 1: Zhipu’s financial profile. H1 2026 is a six-month period and is not directly comparable with the full-year bars. API gross margin recovered to 24.6% , while the shift toward API revenue diluted total gross margin to 26.4% . Sources: Chapter 18C prospectus, 2025 annual report and 2026 interim results^{3–5}.

Listing and market reaction. Zhipu priced at HK\$116.20, raising about HK\$5.0bn after the over-allotment option with HK\$3bn of cornerstone commitments, at an implied valuation near US\$6.7–7.4bn^{3,4}. The listing is strategically remarkable on three counts: Chapter 18C gave a loss-making lab permanent public capital years before profitability; its first-listed-foundation-model status conferred a certification premium; and the stock then re-rated to HK\$1,195 by 31 August 2026 ($\approx +928\%$ versus IPO; realised annualised volatility near 192% ; Figure 7). The company placed 19.78m new shares on 13 July at HK\$1,588, raising HK\$31.375bn net and increasing issued shares from 445.843m to 465.623m⁶. MiniMax, listing one day later at HK\$165, closed 31 August at HK\$349 after a much larger peak-to-trough drawdown; that divergence is used in Section 5.

3 Industry, Macro, and Strategy

Structure and strategy. In Porter terms the foundation-model layer is harsh: supplier power (advanced compute) is high and politically rationed, buyer switching costs are low because open weights commoditise the layer, and the substitutes are rival open-weight models. Zhipu’s answer is leaderboard dominance through *cost disruption*: a mixture-of-experts GLM-5 line (GLM-4.6: $\approx 355\text{B}$ parameters, 32B active, 200K context) that topped coding benchmarks—

GLM-5.1 ranked #1 on SWE-Bench Pro at 58.4%, and GLM-5.2 (June 2026) shipped with MIT open weights and a one-million-token context, followed by GLM-5.3 on 14 August and GLM-5.3-Flash on 26 August—priced below closed competitors, now paired with nascent pricing power (8–17% API price rises with each GLM-5.x release)^{13,15,16}. Branch releases such as GLM-5V-Turbo are catalogued but excluded when they overlap a main GLM window¹⁴. This cadence defines the event taxonomy of Section 5 (Figure 2). The main weakness is self-inflicted: the same cadence commoditises its own product, while cloud-margin pressure and compute dependence cap monetisation.

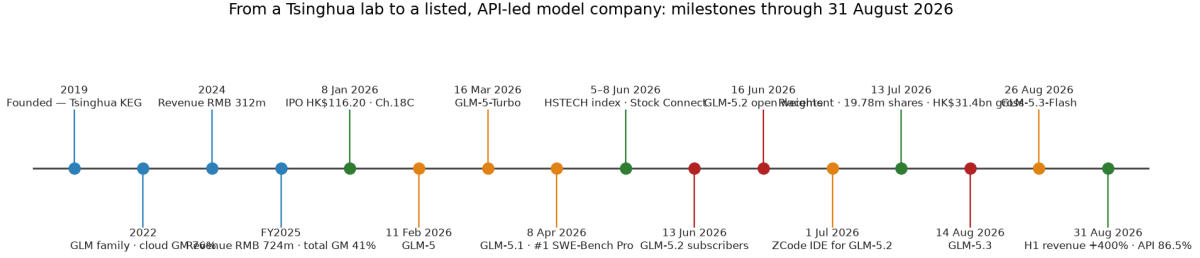


Figure 2: Company timeline from the 2019 founding through the 2026 Chapter 18C listing, July placement, GLM-5.3-Flash rollout and H1 results. Each open-weight launch is the unit of *capability surprise* studied in Section 5 and the primary currency of value for an early-commercial-stage lab; milestones are evenly spaced for legibility. Sources: Chapter 18C prospectus³, official model documentation¹³, placement announcement⁶, and interim results⁵.

The Hong Kong AI-IPO landscape. SenseTime (0020.HK), Fourth Paradigm (6682.HK), and Wenge AI (01956.HK) give local AI context, but their vision, enterprise-ML, and decision-intelligence models differ from a frontier laboratory^{35–37}. Moonshot AI (Kimi) and other Chinese labs may eventually broaden the listed set. For now, *MiniMax is the only direct listed comparable* (Section 4.4).

Macro and flows. Chinese AI is a policy priority (state/SOE demand); U.S. compute-export controls force the architectural efficiency that doubles as Zhipu’s cost edge; and Hong Kong’s market structure sets the marginal buyer. Zhipu joined the Hang Seng Tech Index on 5 June and Southbound Stock Connect on 8 June 2026, drawing about HK\$920m of net southbound buying on its first eligible day³⁸. MSCI then added the H shares to its Emerging Markets and China indexes at the 31 August close³⁹. These flow events matter for Section 5, where we keep them separate from capability and financial-disclosure events.

4 Valuation

4.1 Cost of capital

We estimate the cost of equity with the CAPM. Zhipu’s short post-IPO history makes its own regression beta unstable, so we estimate 60-month monthly betas for global AI-software comparables against SPY through 31 August, unlever them, and round the 1.55 median to $\beta = 1.6$ ⁴⁰. With a risk-free rate of 4.0% and an equity risk premium of 6.0%, $K_e = 4.0\% + 1.6 \times 6.0\% = 13.6\%$; as the firm is effectively all-equity, $WACC \approx 13.5\%$. Throughout we use HK\$7.8/US\$, RMB 7.1/US\$, and 465.623m shares after the July placement. The equity bridge adds **US\$4.289bn of pro-forma net cash and short-term investments**: reported cash plus short-term investments, less bank loans and lease liabilities at 30 June, plus the placement’s net proceeds^{5,6}. This is a transparent model input, not a reported 31 August cash balance; the

interim-results announcement contains no cash-flow statement, so subsequent operating burn cannot yet be observed. Appendix C lists the comparable set and unlevering bridge.

Two caveats remain. First, the beta set consists of mature software firms, while Zhipu resembles a pre-profit, capital-intensive lab; the resulting WACC is generous. Second, fixing WACC at 13.5% through 2035 ignores the risk-premium increase that could follow a break in the valuation loop. Both refinements would lower DCF value. Figure 5 tests rate sensitivity, while the real-options section treats the binary tail.

4.2 Scenario DCF

We project FCFF over 2026–2035 from a **US\$700m FY2026E revenue base**. That figure is our estimate, not company guidance. It starts from reported H1 revenue of RMB 953.9m and implies RMB 4.02bn in H2; the step-up is demanding, though below the August run-rate reported by management after the results^{5,7}. Revenue growth then fades from a scenario-specific 2027 rate to maturity. Operating margin begins at -100% —still a sharp improvement from H1’s -225% —and converges toward a terminal level. The model applies a 15% tax rate, reinvestment through a sales-to-capital ratio of 1.6–2.0, and terminal growth of 4%. When EBIT is negative, cash tax is zero and losses build a simple NOL carryforward; positive EBIT is taxed only after that balance is absorbed. Appendix B reports the full Base-case chain, and Table 1 gives the headline results.

Table 1: Three-scenario DCF (USD unless noted; per-share in HKD). Market row at the 31 August close, before the interim-results announcement.

Scenario	Rev. CAGR '26–35	Term. margin	Equity value	Per share (HK\$)
Bear ($p = 0.35$)	23%	18%	US\$1.5B	25
Base ($p = 0.45$)	31%	28%	US\$4.3B	73
Bull ($p = 0.20$)	47%	35%	US\$13.9B	233
Probability-weighted	—	—	US\$5.3B	88
<i>Market (2026-08-31)</i>	—	—	<i>US\$71.3B</i>	<i>1,195</i>

The probability-weighted intrinsic value is HK\$88 per share—roughly **7.4% of the market price**. The H1 evidence lifts the valuation starting point, but the quoted price still requires a much larger outcome than this conventional cash-flow model.

4.3 Reverse DCF: what the price requires

We invert the model to ask what justifies the US\$71.3bn equity value while holding FY2026E revenue fixed. At WACC 13.5% and a 40% terminal margin, the price requires **about US\$91.9bn of 2035 revenue**: 2027 growth of roughly 148%, followed by a path that still compounds at 71.9% from 2026 to 2035. Figure 3 grids WACC (10–20%) against terminal margin (20–50%). Even the most favourable cell (10%/50%) requires about US\$28bn, while 15%/40% requires US\$126bn. The reverse DCF is therefore less a target price than a statement of the right-tail outcome already embedded in the quote.

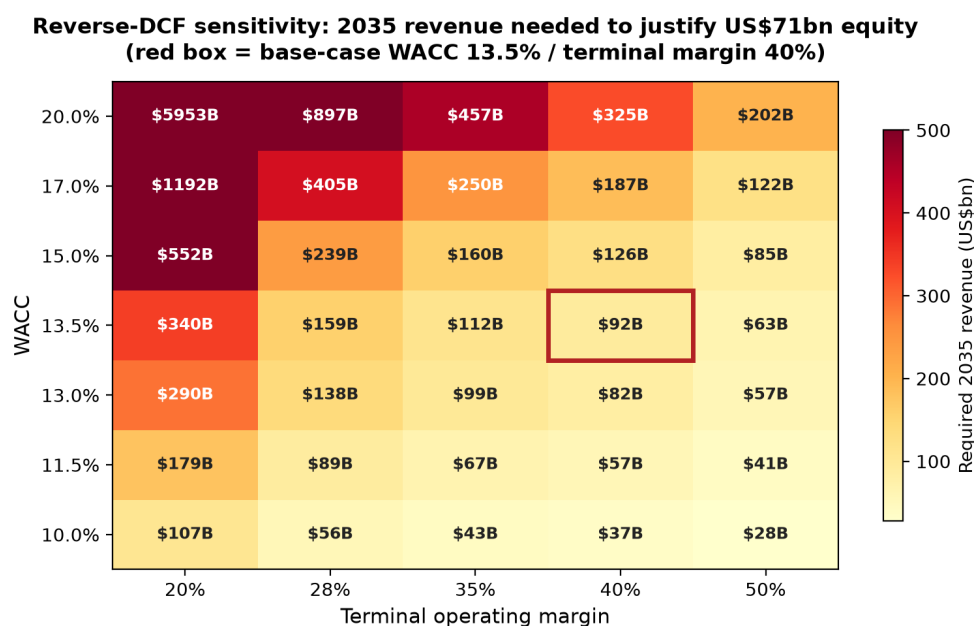


Figure 3: Reverse-DCF sensitivity: 2035 revenue (US\$bn) required to justify the US\$71.3bn equity value across WACC and terminal-margin assumptions. The red box marks the base case (13.5%/40%). Data: Table 9 in Appendix D.

4.4 Relative valuation

We divide each listed lab's 31 August equity value by LTM revenue through 2026H1. Zhipu's RMB 1,487.3m (FY2025 less H1 2025 plus H1 2026), or US\$209.5m at RMB7.1/US\$, gives **340.5×**. MiniMax's US\$15.63bn equity value and US\$165.2m LTM revenue give **94.6×**. Private transactions imply **34.1×** for OpenAI, **20.5×** for Anthropic, and **39.0×** for Mistral^{41–44}. This private range, with a 34.1× median, is the football-field transaction range. Their revenue run-rates apply separately to Zhipu's FY2026E estimate, giving HK\$240–457 per share. MiniMax's LTM multiple applied to Zhipu's LTM revenue gives HK\$332.

MiniMax's 26 August results anchor the peer denominator. H1 revenue rose 283.1% to US\$116.6m and gross margin reached 17.9%, while adjusted net loss doubled to US\$293.0m. Its US\$165.2m LTM revenue equals FY2025 less H1 2025 plus H1 2026, rather than annualised H1 revenue⁴³.

SenseTime, Phancy, and Wenge are Hong Kong adjacent-AI references; Palantir, Cloudflare, and Snowflake represent commercialisation^{45,46}. Different economics make both groups context only. On the matched basis, Zhipu's multiple is 3.60 times MiniMax's, so the peer supports far less of the quote (Figure 4).

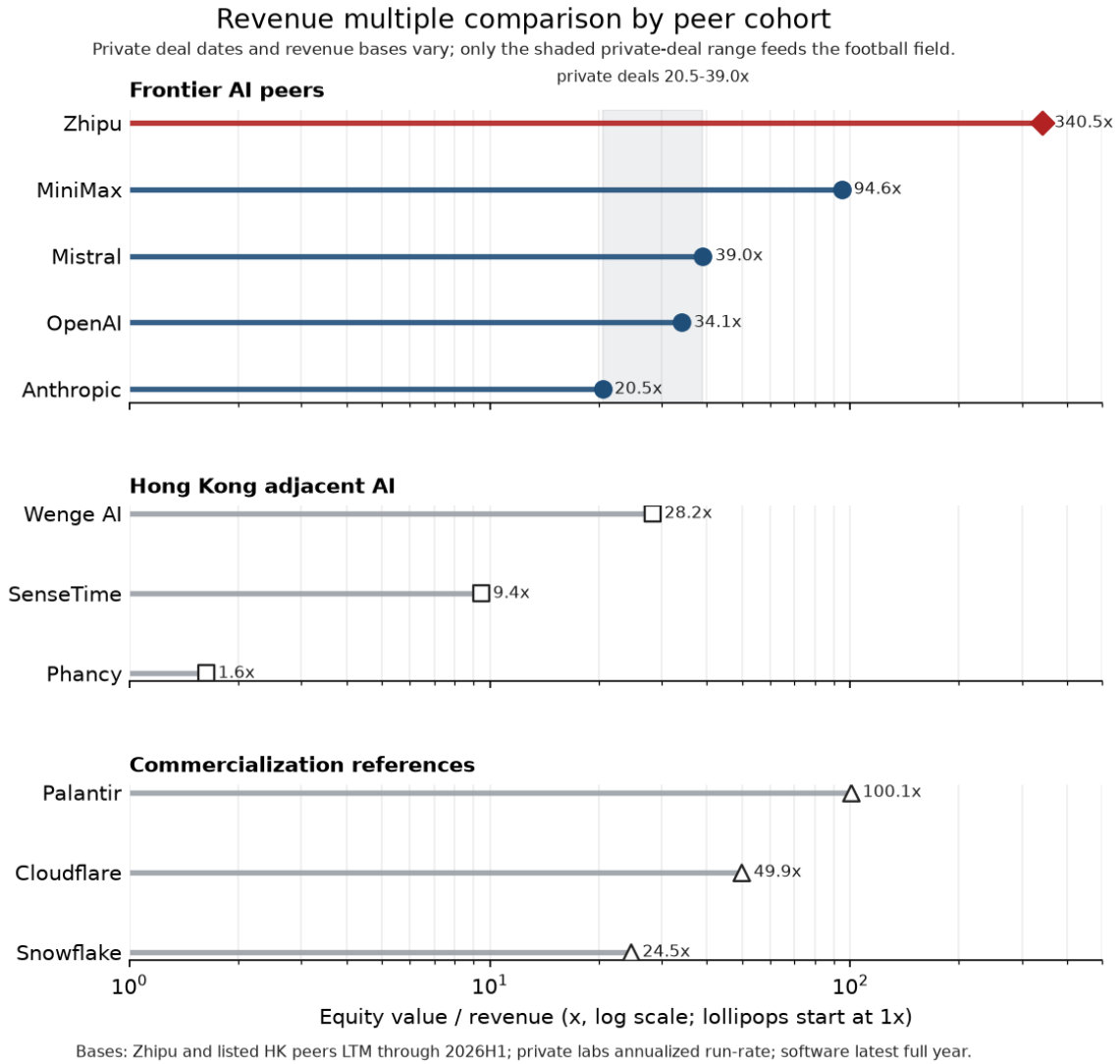


Figure 4: Layered equity value / revenue multiples (log scale). The chart shows core frontier peers, Hong Kong adjacent-AI firms, and commercialisation references in separate groups. Only the shaded private-deal range from 20.5 $\times$ to 39.0 $\times$ feeds the football field. Zhipu, MiniMax, and the Hong Kong listed peers use LTM revenue through 2026H1; each other revenue period is recorded in `data/valuation_comps.csv`.

4.5 A real-options reading

The reconciliation is that Zhipu is best valued not as a discounted cash-flow stream but as a *call option on a winner-take-all AGI outcome*^{47,48}. The DCF base case is the option's modest intrinsic value; the rest is time value on a fat right tail, amplified by roughly 192% annualised volatility and a long horizon. The reverse DCF describes the state needed for that option to pay off, while Section 5 tests how capability news changes its perceived probability.

A practitioner lens: reflexivity. This also fits Feng Liu's reverse-investing interpretation of Soros's *reflexivity*⁴⁹. A high valuation is an input as well as an output: cheap equity capital finances compute and talent, which can improve models and support the valuation. The July placement makes that loop tangible. It remains fragile because the firm does not yet fund itself; if capability surprises or monetisation stall, the same loop can reverse.

4.6 Sensitivity and synthesis

The conclusion is not an artefact of any single assumption. Figure 5 tornadoes the Base case against its main drivers. One-at-a-time changes put the Base case between roughly HK\$43 and HK\$104: useful for locating model risk, but insufficient to bridge the gap to market. Figure 6 draws the whole picture together. Scenario DCF spans HK\$25–233, the 20.5–39.0 $\times$ private-deal range gives HK\$240–457, and MiniMax’s 94.6 $\times$ LTM multiple gives about HK\$332 when applied to Zhipu’s LTM revenue. That anchor is 72.2% below Zhipu’s HK\$1,195 close. Public markets value both listed labs richly, but Zhipu still carries a 3.60-fold multiple premium to MiniMax, making its price unusually dependent on a durable right-tail option.

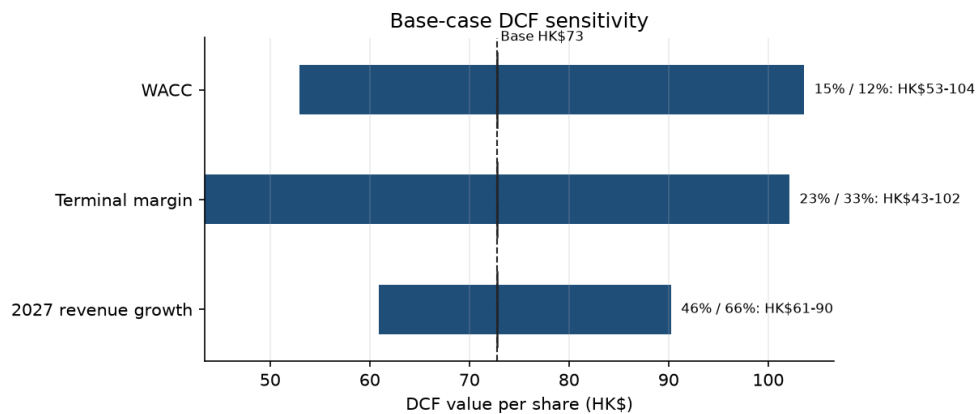


Figure 5: DCF sensitivity (Base case, per share). Bars span each driver’s low/high setting around the HK\$73 base; no one-at-a-time input change moves value near the market price.

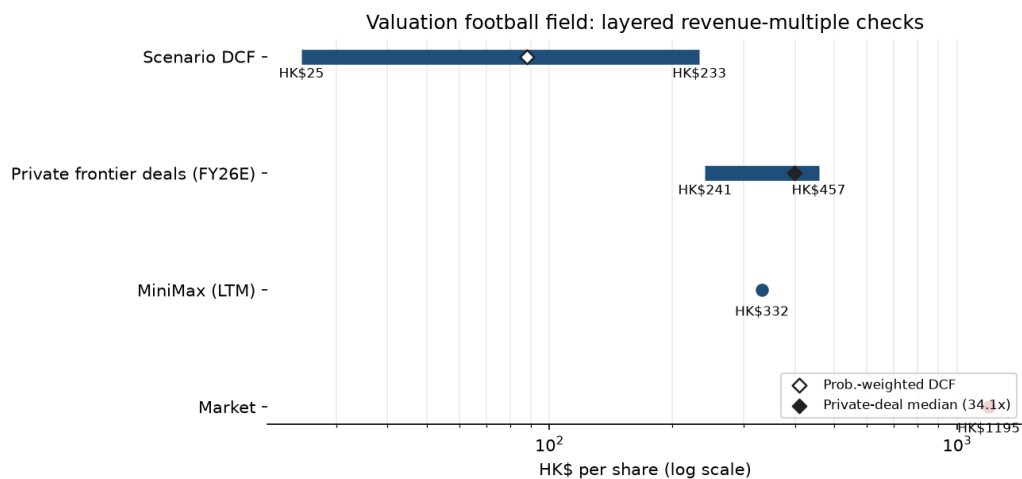


Figure 6: Valuation football field (log scale). The private-deal range and MiniMax’s public LTM multiple are shown separately; the diamonds mark probability-weighted DCF and the private-deal median.

5 Capability Surprise: An Event Study

Hypothesis. If the firm is priced on its capability trajectory, then capability information should move the price. Under semi-strong efficiency⁵⁰ the reaction should be immediate and complete; any systematic *post-capability-announcement drift* (PCAD)—the analogue of the classic post-earnings-announcement drift^{10,11}—would indicate inefficiency.

Method. Event dates are identified *independently* from official model-release announcements and technical documentation^{13,17}, not from the price series. On daily prices^{1,2} we compute mean-adjusted abnormal returns and CARs over three windows—leakage $[-5, -1]$, reaction $[0, +1]$, and drift $[+2, +10]$ —following standard event-study practice⁵¹ and the surprise-event design of Guosen Securities¹². For each event the expected daily return is Zhipu’s mean return over the pre-event estimation window $[-20, -6]$, and the abnormal return is $AR_t = R_t - \bar{R}_{[-20, -6]}$. Figure 8 then serves as a *descriptive cross-check*: several of the largest positive-return days coincide with these independently dated capability events.

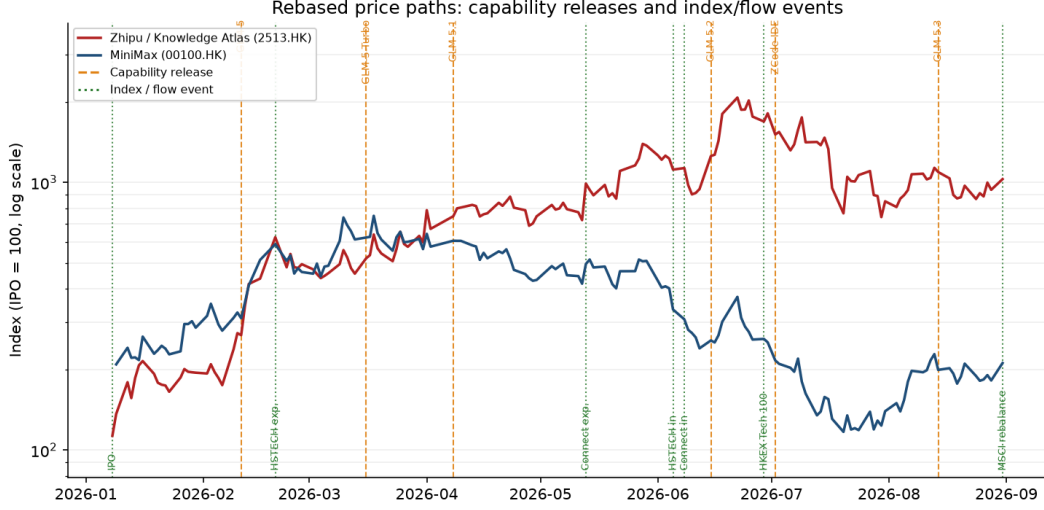


Figure 7: Rebased prices (IPO = 100, log scale, through 2026-08-31). Orange dashed lines mark capability releases (GLM-5 through GLM-5.3); green dotted lines mark index/flow catalysts (IPO, the 20 Feb and 13 May expectation spikes, the 5–8 June Hang Seng TECH inclusion and Stock Connect access, and the 31 August MSCI rebalance). Same sector, divergent paths: Zhipu re-rates on successive SOTA releases while MiniMax de-rates from its peak.

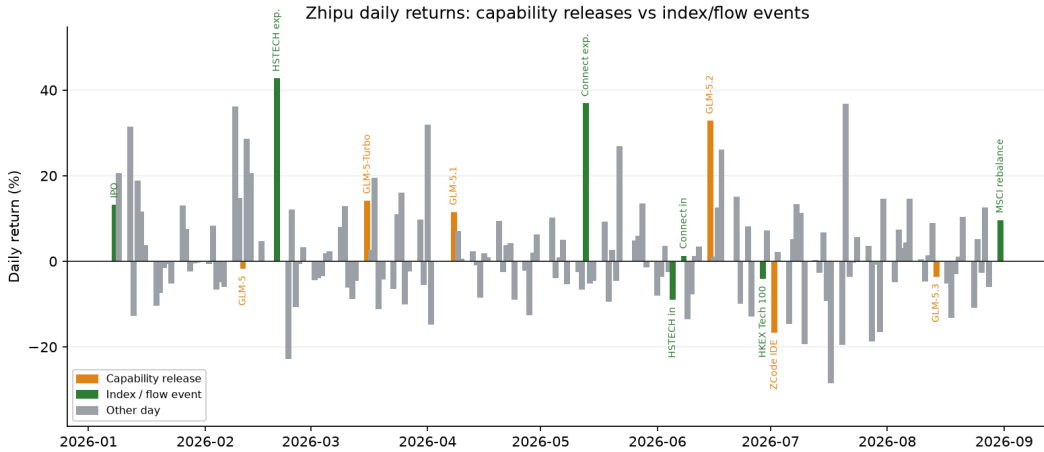


Figure 8: Descriptive cross-check: Zhipu’s daily returns with capability-release days (orange) and index/flow days (green) highlighted. Several of the largest up-days coincide with independently dated GLM releases, but the two biggest—+43% on 20 Feb and +37% on 13 May—are index/flow events, not model releases. (Event dates are taken from official announcements and index/Connect schedules, not from this chart.)

Results. Averaged across the five GLM events, the reaction window is +13.7% (the market prices capability quickly) and the drift averages +4.6% but is strongly *bimodal* (Table 2).

The drift sign depends on the *type* of release: a genuine SOTA leap (GLM-5.2) continues upward, whereas incremental upgrades (GLM-5-Turbo, GLM-5.1) reverse. MiniMax M2.5 drew a +24.2% launch reaction with little subsequent drift, whereas M2.7 and M3 both drew negative reaction/drift profiles. That contrast is consistent with *capability surprise, not sector membership, driving the cross-section* (Figures 9–10).

Table 2: Cumulative abnormal returns around capability events (mean-adjusted, %; data through 31 August 2026).

Event	Day 0	Reaction [0, +1]	Drift [+2, +10]	Reading
GLM-5	2026-02-11	+22.5	+24.7	under-reaction
GLM-5-Turbo	2026-03-16	+7.7	−19.3	reversal
GLM-5.1	2026-04-08	+13.8	−14.2	over-reaction
GLM-5.2	2026-06-15	+30.8	+28.2	strong under-reaction
GLM-5.3	2026-08-14	−6.5	+3.8	anticipated / delayed reaction
MiniMax M2.5	2026-02-12	+24.2	−2.1	front-loaded reaction
MiniMax M2.7	2026-03-18	−5.5	−49.1	de-rate
MiniMax M3	2026-06-01	−21.1	−40.8	failed catalyst
<i>Large-tech AI comparators</i>				
Baidu ERNIE 5.0	2026-02-06	−0.4	−15.8	de-rate
Baidu ERNIE 5.1	2026-05-11	−5.0	−13.6	de-rate
Alibaba Qwen3.5	2026-02-16	−4.5	−10.9	de-rate
Alibaba Qwen3.6	2026-04-16	+6.5	−5.5	reversal
Alibaba Qwen3.7-Max	2026-05-21	−3.4	−2.1	de-rate
Alibaba Qwen3.7-Plus	2026-06-01	+8.6	−15.9	reversal
Alibaba Qwen3.8-Max	2026-08-03	+5.3	−12.8	reversal
Tencent Hunyuan 3.0 preview	2026-04-22	−4.2	−1.3	de-rate
Tencent Hy3	2026-07-06	+8.2	+9.8	under-reaction
Meituan LongCat-2.0	2026-06-30	+4.8	+17.3	under-reaction
Google Gemini 3.1 Pro	2026-02-19	+4.0	−4.0	reversal
Google Gemini 3.5/Omni	2026-05-19	−4.0	−16.7	de-rate
Google Gemini 3.6 Flash	2026-07-21	−2.3	+13.0	delayed reaction
Google Gemini 3.7 Flash	2026-08-13	+0.9	−0.7	muted
Meta Muse Spark	2026-04-08	+11.5	+18.2	under-reaction
Meta Muse Spark 1.1	2026-07-09	+11.1	−7.6	reversal
Average (16 large-tech)		+2.3	−3.0	
Average (5 GLM)		+13.7	+4.6	

Mean-adjusted abnormal return: $AR_t = R_t - \bar{R}_{[-20, -6]}$, where $\bar{R}_{[-20, -6]}$ is the average raw return over event days −20 through −6. Comparator dates are pinned to primary-source release announcements or model cards^{17–34}. Weekend announcements use the next trading day. OpenAI and xAI releases remain in the audit catalogue, but Microsoft and Tesla returns are excluded because neither listed company is the model issuer.

Same-week Qwen3.8 variants are catalogued but not double-counted after the 3 August flagship launch.

Qwen3.8-Flash (26 August), Gemini Transcribe/Omni 1.1 (26–27 August), Tencent Hy4 (28 August), GLM-5.3-Flash, the two H1 disclosures, and MSCI rebalance lack a complete post-event window at the cutoff and remain outside the table. MiniMax H3 and Music 3.0 remain catalogued but are excluded because audio-video and music generation are not comparable with the text, coding, and agent capability sample.

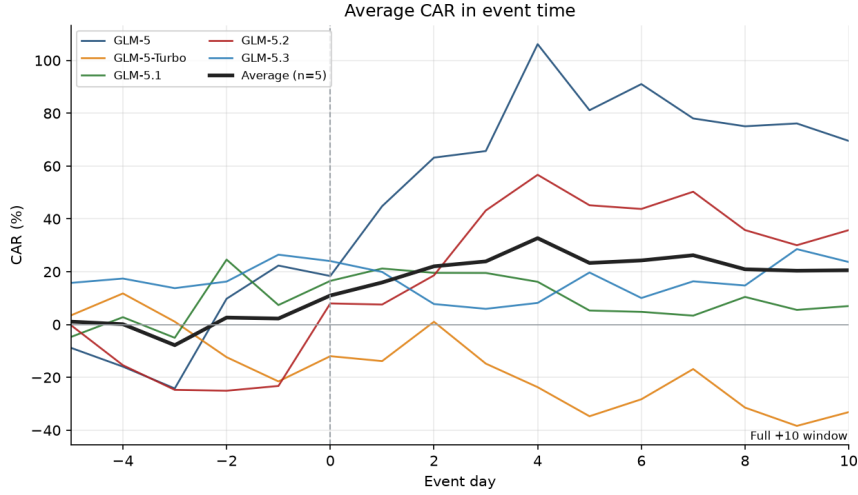


Figure 9: Average CAR in event time across the five GLM events (mean-adjusted, $[-20, -6]$ estimation window). The average is plotted where all five GLM events have data through the full $[+2, +10]$ drift window.

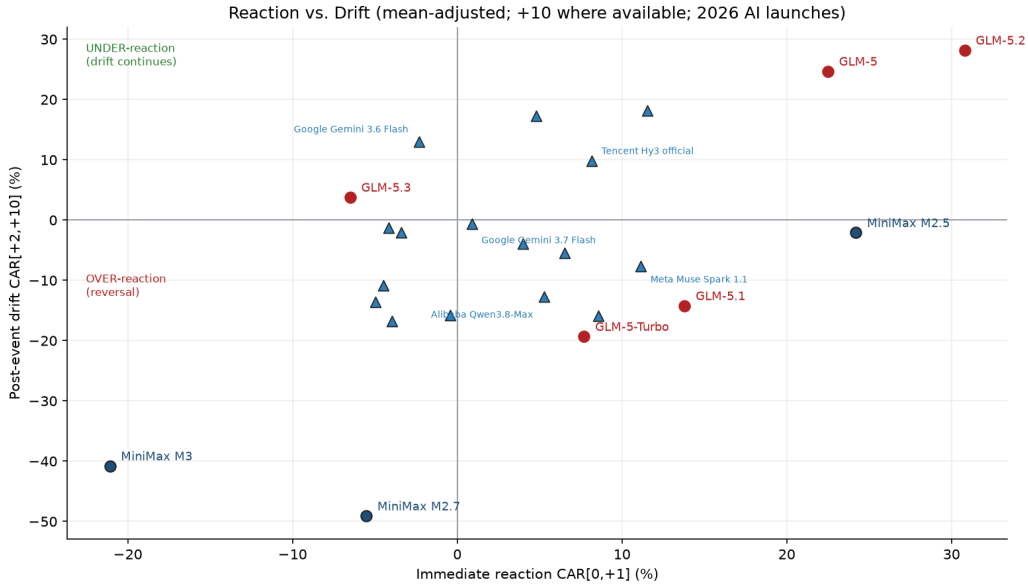


Figure 10: Reaction versus drift. SOTA leaps sit in the under-reaction quadrant; incremental upgrades in the over-reaction quadrant. Triangle markers show large-tech AI comparators.

5.1 Not every spike is capability: flow events

Two of the largest blind-detected up-days do *not* map to a model release: $+43\%$ on 2026-02-20 and $+37\%$ on 2026-05-13. Both coincide with *index/flow* catalysts—rising expectations of Hang Seng Tech inclusion and Stock Connect access (realised 5 and 8 June). Separating capability events from flow events matters: the CARs above are driven by the former, while the latter explain part of the raw volatility and warn against attributing every move to model quality. The same discipline matters at the new cutoff. Zhipu rose 9.6% on 31 August as MSCI’s index changes became effective after the close³⁹; the H1 results were released only after that close⁵. The financial disclosure can update valuation inputs, but it cannot explain that day’s return and has no post-announcement window yet, so we exclude it from the CAR sample.

5.2 Robustness: peer-adjusted and block-bootstrap checks

Because Zhipu and MiniMax share sector and macro shocks, we re-run the study with *peer-adjusted* returns (benchmark = MiniMax’s same-day return). Table 3 shows the reaction result is highly robust (+13.7% vs +15.3%), while the average drift *strengthens* (+4.6% $\rightarrow$ +16.8%): once the falling sector is removed, all five events drift positively, and GLM-5.1’s apparent reversal flips to continuation (−14.2% $\rightarrow$ +12.9%). The under-reaction pattern is reinforced, not weakened, by peer adjustment.

An expanded event catalogue guards against cherry-picking by recording screened releases, financial disclosures, flow catalysts, listings, and candidates. Twenty-nine day-level, clean-window events enter the CAR panel; exclusions carry explicit reasons. Capability reactions remain positive, while flow events show weaker reactions but stronger drift, supporting their separate treatment.

Table 3: Robustness: mean-adjusted vs peer-adjusted CAR (%).

Event	Mean-adjusted		Peer-adjusted	
	React [0, 1]	Drift [2, 10]	React [0, 1]	Drift [2, 10]
GLM-5	+22.5	+24.7	+17.2	+13.3
GLM-5-Turbo	+7.7	−19.3	+14.6	+19.2
GLM-5.1	+13.8	−14.2	+13.4	+12.9
GLM-5.2	+30.8	+28.2	+28.7	+36.8
GLM-5.3	−6.5	+3.8	+2.4	+1.8
Average	+13.7	+4.6	+15.3	+16.8

Mean-adjusted uses Zhipu’s pre-event average return over $[-20, -6]$; peer-adjusted uses MiniMax’s same-day return as the benchmark. All five GLM drift windows are complete through +10.

Block-bootstrap check. With only five events, a parametric t -test would create false precision. We therefore draw 20,000 non-event peer-adjusted return blocks with `numpy.random.choice`, preserving the two-day reaction and nine-day drift window lengths, and compare the actual five-event mean CAR with the bootstrap mean distribution (Figure 11). The actual peer-adjusted reaction (+15.3%) sits above the 95% null interval ($[-14.0, +12.6]\%$) and at the 99.2nd percentile. Drift (+16.8%) is positive but inside its wider null interval ($[-22.8, +25.5]\%$), at percentile 88.0. Thus the block bootstrap supports a strong immediate capability reaction and only directional, not statistically conclusive, continuation.

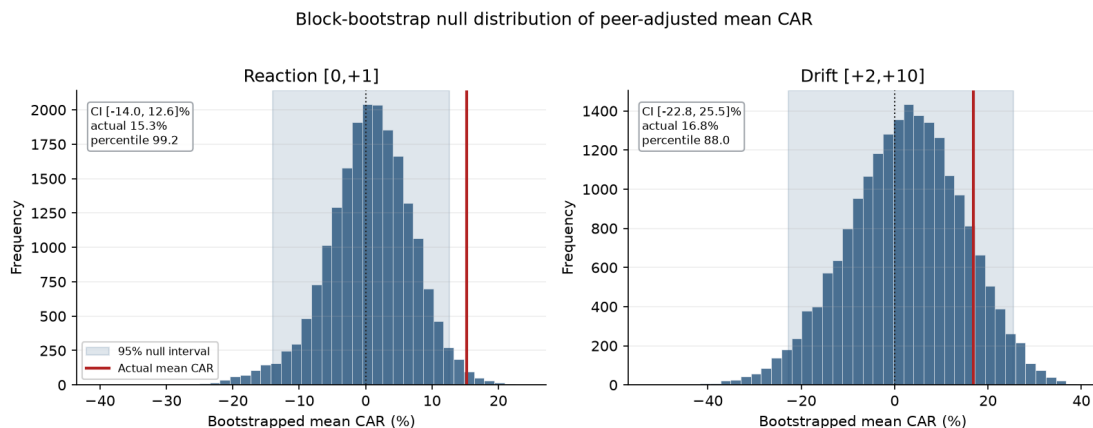


Figure 11: Block-bootstrap null distributions for peer-adjusted mean CAR. Each draw samples five non-event return blocks with replacement using `numpy.random.choice`; block lengths match the actual event windows. Shaded areas are percentile 95% null intervals, and red lines mark the actual five-event means.

Honesty box. This is *preliminary, diagnostic* evidence consistent with PCAD, not a proven anomaly: $n = 5$ single-firm events (plus two direct peer capability events) over seven months, and drift windows can overlap competing-lab news. No statistical significance is claimed—five events make an interesting undergraduate case, not a general law. The natural extensions are a wider, multi-lab event panel and an NLP-scored surprise magnitude.

6 Conclusion and Implications

H1 revenue reached RMB 953.9m and API gross margin turned positive, but operating loss still exceeded twice revenue. Our probability-weighted DCF is HK\$88 per share and private transactions imply HK\$240–457, against a HK\$1,195 pre-results close. MiniMax’s public multiple implies only HK\$332 on a matched LTM basis. Listed scarcity affects both firms, but Zhipu still trades at 3.60 times MiniMax’s revenue multiple. The reverse DCF states the wager—about US\$91.9bn of 2035 revenue.

Capability surprises are priced quickly and discriminately, while index flows explain part of the remaining volatility. On 31 August, MSCI rebalancing could affect the close; results released afterward could not. The verdict is therefore **capability momentum priced as a scarce public option**, not a casual “bubble” label. The main limitations are the small event sample, no post-results return window, and an uncertain path from API volume to durable cash flow.

A Financial Statements Summary

From the Chapter 18C prospectus, 2025 annual report and H1 2026 interim-results announcement (RMB; USD at ≈ 7.1).

Table 4: Income-statement summary (RMB millions unless noted).

	FY2022	FY2023	FY2024	FY2025
Revenue	57.4	124.5	312.4	724.3
Total gross margin	54.6%	64.6%	56.3%	41.0%
Cloud gross margin	76.1%	31.0%	3.4%	18.9%
Net loss	143.7	788.0	2,958.0	4,718.2

Table 5: H1 operating scorecard (RMB millions unless noted).

	H1 2025	H1 2026	Change
Revenue	190.9	953.9	+399.7%
Open-platform and API revenue	29.1	825.2	+2,735.7%
Total gross margin	50.0%	26.4%	−23.6pp
API gross margin	−0.4%	24.6%	+25.0pp
Operating loss	1,899.2	2,146.6	+13.0%
Adjusted net loss	1,752.0	1,964.1	+12.1%

Table 6: Balance-sheet and placement bridge (RMB millions at 30 June 2026 unless noted).

Item	Amount
Cash and cash equivalents	3,993.7
Short-term investments at FVPL	506.1
Bank loans	2,224.8
Lease liabilities	379.4
Total equity	4,429.6
Shares outstanding after placement	465.623m
Placement price / net proceeds	HK\$1,588 / HK\$31,375.0m
Model pro-forma net cash and investments	US\$4,289.4m

The central operating fact is a genuine commercial inflection paired with continuing cash intensity. API revenue and unit gross margin improved sharply, but mix shift compressed total gross margin and operating loss still widened. The July placement removed the immediate financing constraint; it did not establish a reported 31 August cash balance or solve the underlying path to positive free cash flow.

B Base-Case DCF Projection

Base-case chain (US\$m unless noted), reproducing the live workbook: WACC = 13.5%, statutory tax = 15%, terminal $g = 4\%$, sales-to-capital = 1.8, 465.623m shares, HK\$7.8/US\$. Revenue starts from the model's US\$700m FY2026E and grows to US\$8.0bn by 2035, with annual growth fading from 56% in 2027 to 8% in 2035. Operating margin ramps $-100\% \rightarrow 28\%$. Cash tax is zero while EBIT is negative; losses accumulate as NOLs and shelter future positive EBIT until exhausted.

Table 7: Base-case projection (US\$ millions unless noted).

Year	Revenue	Growth	EBIT margin	Cash tax	NOL end	NOPAT	Reinvest	FCFF
2026	700	586%	-100.0%	0	700	-700	332	-1,032
2027	1,092	56%	-85.8%	0	1,637	-937	218	-1,155
2028	1,638	50%	-71.6%	0	2,809	-1,172	303	-1,475
2029	2,359	44%	-57.3%	0	4,161	-1,352	400	-1,753
2030	3,255	38%	-43.1%	0	5,564	-1,403	498	-1,901
2031	4,297	32%	-28.9%	0	6,806	-1,241	579	-1,820
2032	5,414	26%	-14.7%	0	7,600	-794	621	-1,415
2033	6,497	20%	-0.4%	0	7,629	-29	602	-630
2034	7,406	14%	13.8%	0	6,608	1,020	505	515
2035	7,999	8%	28.0%	0	4,369	2,240	329	1,910

Terminal value (2035) = $1,910 \times 1.04 / (0.135 - 0.04)$ = US\$20,914m; PV of TV = US\$5,895m.

Enterprise value = US\$55m; + pro-forma net cash and investments US\$4,289m = equity US\$4,344m = **HK\$73/share**.

C Bottom-Up Beta Bridge

The bottom-up beta uses public software and AI-infrastructure comparables with observable market betas and capital structures. Levered betas use 60 month-end returns through 31 August 2026; market value is that day’s close times the latest disclosed shares, and debt comes from the latest pre-cutoff SEC filing. We unlever as $\beta_U = \beta_L/[1 + (1 - T)D/E]$, using a 21% tax rate, market capitalisation as equity value, and total debt as debt⁴⁰. Because Zhipu is effectively all-equity, we round the 1.55 median to $\beta = 1.6$ as a modest early-stage risk buffer.

Table 8: Comparable-company beta bridge (60-month beta window ended 31 August 2026).

Comparable	Levered beta	Debt/equity	Unlevered beta
Salesforce	1.19	20.0%	1.03
Palantir	1.61	0.0%	1.61
Snowflake	1.33	2.4%	1.30
Datadog	1.48	1.5%	1.46
MongoDB	1.55	0.2%	1.55
Cloudflare	1.66	4.4%	1.60
C3.ai	2.11	3.4%	2.05
Median	—	—	1.55

D Reverse-DCF Sensitivity Grid

Table 9 reports the 2035 revenue (US\$bn) required to justify the US\$71.3bn equity value across a grid of WACC and terminal-margin assumptions, underlying Figure 3.

Table 9: Reverse-DCF sensitivity: required 2035 revenue (US\$bn) to justify US\$71.3bn equity.

WACC	Terminal operating margin				
	20%	28%	35%	40%	50%
10.0%	107	56	43	37	28
11.5%	179	89	67	57	41
13.0%	290	138	99	82	57
13.5%	340	159	112	92	63
15.0%	552	239	160	126	85
17.0%	1,192	405	250	187	122
20.0%	5,953	897	457	325	202

Bold row = base-case WACC. All figures hold FY2026E revenue at US\$700m and solve the 2027 growth rate, after which growth fades linearly to 8% in 2035. They also assume $g = 4\%$, sales-to-capital 1.8 and a 15% tax rate with NOL carryforwards.

References

- [1] Tushare. (2026). *Tushare Pro financial data API: Hong Kong daily quotations* (hk_daily) [Data set]. <https://tushare.pro>
- [2] yfinance. (2026). *Yahoo Finance market data downloader for Python* [Data set]. GitHub. <https://github.com/ranaroussi/yfinance>
- [3] Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI). (2025). *Global offering prospectus (Stock code 2513)* [Chapter 18C listing document]. HKEXnews. <https://www1.hkexnews.hk/listedco/listconews/sehk/2025/1230/2025123000017.pdf>
- [4] Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI). (2026). *2025 annual report (Stock code 2513)*. HKEXnews. <https://www1.hkexnews.hk/listedco/listconews/sehk/2026/0419/2026041900085.pdf>
- [5] Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI). (2026). *Interim results for the six months ended 30 June 2026* [Unaudited announcement released after market close on 31 August 2026]. HKEXnews. <https://www1.hkexnews.hk/listedco/listconews/sehk/2026/0831/2026083101539.pdf>
- [6] Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI). (2026). *Placing of new H shares under the general mandate* [13 July 2026]. HKEXnews. <https://www1.hkexnews.hk/listedco/listconews/sehk/2026/0713/2026071300662.pdf>
- [7] Caixin. (2026). *Zhipu AI’s first-half revenue approaches RMB 1bn; management discusses August annualised revenue run-rate* [31 August 2026]. <https://companies.caixin.com/2026-08-31/102480428.html>
- [8] Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI). (2026). *Poll results of the 2025 annual general meeting held on 22 June 2026 (Stock code 2513)*. HKEXnews. https://www.hkexnews.hk/listedco/listconews/sehk/2026/0622/2026062202116_c.pdf
- [9] MiniMax Inc. (2025). *Global offering prospectus (Stock code 100)*. HKEXnews. <https://www1.hkexnews.hk/listedco/listconews/sehk/2025/1231/2025123100025.pdf>
- [10] Ball, R., & Brown, P. (1968). An empirical evaluation of accounting income numbers. *Journal of Accounting Research*, 6(2), 159–178.
- [11] Bernard, V. L., & Thomas, J. K. (1989). Post-earnings-announcement drift: Delayed price response or risk premium? *Journal of Accounting Research*, 27, 1–36.
- [12] Guosen Securities Economic Research Institute. (2020). *A complete guide to earnings-surprise investing* [Special report on financial engineering]. Guosen Securities.
- [13] Z.ai. (2026). *GLM-5 and GLM-5.1 model cards and technical reports* [Architecture, context length, and benchmark results]. Hugging Face model cards and official technical blog. <https://huggingface.co/zai-org/GLM-5>; <https://huggingface.co/zai-org/GLM-5.1>
- [14] GLM-V Team. (2026). *GLM-5V-Turbo: Toward a native foundation model for multimodal agents* (arXiv:2604.26752). arXiv. <https://arxiv.org/abs/2604.26752>
- [15] Z.ai. (2026). *GLM-5.3: From Vibe Coding to Agentic Engineering* [14 August 2026]. <https://z.ai/blog/glm-5.3>

- [16] Z.ai. (2026). *GLM-5.3-Flash: Faster Agentic Coding* [26 August 2026]. <https://z.ai/blog/glm-5.3-flash>
- [17] MiniMax AI. (2026). *MiniMax M3 model card and MiniMax Sparse Attention technical report* [Model card and technical report]. Hugging Face and arXiv. <https://huggingface.co/MiniMaxAI/MiniMax-M3>; <https://arxiv.org/abs/2606.13392>
- [18] MiniMax AI. (2026). *MiniMax M2.5: Built for Real-World Productivity* [12 February 2026]. Official M2.5 release
- [19] MiniMax AI. (2026). *MiniMax H3* [31 July 2026] and *MiniMax Music 3.0* [13 August 2026]. Official H3 release; Official Music 3.0 release
- [20] Baidu ERNIE Team. (2026). *ERNIE 5.0 technical report* (arXiv:2602.04705). arXiv. <https://arxiv.org/abs/2602.04705>
- [21] Baidu ERNIE Team. (2026). *ERNIE 5.1 officially released* [9 May 2026]. <https://ernie.baidu.com/blog/posts/ernie-5.1-0508-release/>
- [22] Qwen Team. (2026). *Qwen3.5 open-model release record* [16 February 2026]. <https://github.com/QwenLM/Qwen3.8>
- [23] Qwen Team. (2026). *Qwen3.6 open-model release record* [April 2026]. <https://github.com/QwenLM/Qwen3.8>
- [24] Alibaba Cloud and Qwen Team. (2026). *Qwen3.8-Max launch and Qwen3.8 open-model release record* [August 2026]. <https://modelstudio.alibabacloud.com/>; <https://github.com/QwenLM/Qwen3.8>
- [25] Tencent. (2026). *Hy3 official release* [6 July 2026] and *Hy4 preview release* [28 August 2026]. Tencent Hy3 release; Tencent Hy4 preview release
- [26] Meituan Technical Team. (2026). *LongCat-2.0 official release* [30 June 2026]. <https://tech.meituan.com/2026/06/30/LongCat2.0.html>
- [27] Google. (2026). *Google I/O 2026: News and announcements* [19 May 2026]. <https://blog.google/innovation-and-ai/technology/developers-tools/google-io-2026-collection/>
- [28] Google. (2026). *Introducing Gemini 3.6 Flash* [21 July 2026] and *Gemini 3.7 Flash* [13 August 2026]. Google Gemini 3.6 Flash release; Google Gemini 3.7 Flash release
- [29] Meta Superintelligence Labs. (2026). *Muse Spark Safety & Preparedness Report* (arXiv:2606.12429). <https://arxiv.org/abs/2606.12429>
- [30] Meta Superintelligence Labs. (2026). *Introducing Muse Spark 1.1* [9 July 2026]. <https://ai.meta.com/blog/introducing-muse-spark-meta-model-api/>
- [31] OpenAI. (2026). *Introducing GPT-5.5* [23 April 2026]. <https://openai.com/index/introducing-gpt-5-5/>
- [32] OpenAI. (2026). *GPT-5.6* [9 July 2026] and *GPT-5.6—August updates* [6 August 2026]. <https://openai.com/index/gpt-5-6/>; <https://deploymentsafety.openai.com/gpt-5-6-august-update>
- [33] Grok. (2026). *Grok chatbot release record: Grok 4.3 Beta* [17 April 2026]. <https://grok.com/>

- [34] SpaceXAI. (2026). *Introducing Grok 4.5* [20 July 2026] and *Introducing Grok 4.6* [12 August 2026]. <https://x.ai/news/grok-4-5>; <https://x.ai/news/grok-4-6>
- [35] SenseTime Group Inc. (2021). *Global offering prospectus (Stock code 20)*. HKEXnews. <https://www1.hkexnews.hk/listedco/listconews/sehk/2021/1207/2021120700017.pdf>
- [36] Beijing Fourth Paradigm Technology Co., Ltd. (2023). *Global offering prospectus (Stock code 6682)*. HKEXnews. <https://www1.hkexnews.hk/listedco/listconews/sehk/2023/0918/2023091800009.pdf>
- [37] Wenge Technology Co., Ltd. (2026). *Global offering prospectus (Stock code 1956)*. HKEXnews. <https://www1.hkexnews.hk/listedco/listconews/sehk/2026/0617/2026061700025.pdf>
- [38] Hang Seng Indexes Company Limited. (2026). *Index review results and constituent-change announcements*. Hang Seng Indexes Co.
- [39] MSCI. (2026). *MSCI Equity Indexes August 2026 Index Review* [Changes effective after the close of 31 August 2026]. <https://ir.msci.com/node/22896/pdf>
- [40] Nasdaq and U.S. Securities and Exchange Commission. (2026). *Monthly price histories through 31 August 2026 and latest pre-cutoff company filings*. Row-level sources and calculation notes are recorded in `data/comps_beta_bridge_input.csv`. <https://www.nasdaq.com>; <https://www.sec.gov>
- [41] OpenAI. (2026). *Accelerating the next phase of AI* [31 March 2026]; Reuters. (2026). *OpenAI tops US\$25bn annualised revenue*. <https://openai.com/index/accelerating-the-next-phase-ai/>; <https://finance.yahoo.com/news/openai-tops-25-billion-annualized-033836274.html>
- [42] Anthropic. (2026). *Announcing our Series H* [28 May 2026]. <https://www.anthropic.com/news/series-h>
- [43] MiniMax Group Inc. (2026). *Interim results for the six months ended 30 June 2026*. HKEXnews. <https://www1.hkexnews.hk/listedco/listconews/sehk/2026/0826/2026082600680.pdf>
- [44] Mistral AI. (2025). *Series C financing*; Le Monde. (2025). *Mistral AI: incarnation de l'intelligence artificielle*. <https://mistral.ai/news/mistral-ai-raises-1-7-b-to-accelerate-technological-progress-with-ai/>; https://www.lemonde.fr/economie/article/2025/09/26/mistral-ai-incarnation-de-l-intelligence-artificielle_6643087_3234.html
- [45] SenseTime Group, Phancy Group, and Wenge Technology. (2026). *2025 annual and 2026 interim reports*. HKEXnews. <https://www1.hkexnews.hk>; calculations and document links in `data/valuation_comps.csv`.
- [46] Palantir Technologies, Cloudflare, and Snowflake. (2026). *2025/FY2026 annual reports and 31 August 2026 market-capitalisation snapshots*. SEC, company investor relations, and public market data; document links in `data/valuation_comps.csv`.
- [47] Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- [48] Damodaran, A. (2018). *The dark side of valuation: Valuing young, distressed, and complex businesses* (3rd ed.). Pearson Education.

- [49] Feng, L. (2017). *Collected investment essays*. Gao Yi Asset Management.
- [50] Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- [51] MacKinlay, A. C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1), 13–39.